

Transcript: ASHA Voices: Researchers Discuss Speech in Noisy Environments, Spatial Awareness, and Bilateral Cochlear Implants



GRAY (Narration):

This is ASHA Voices. I'm J.D. Gray.

The conversations you'll hear today will introduce you to researchers asking questions about the information we receive in sound. And both guests have something to say about both speech in noise and spatial awareness.

Every year on the podcast, we bring you conversations from the speakers at ASHA's Research Symposium on Hearing, a part of the ASHA Convention. Among the speakers is Matt Goupell. He talks to us about the use of bilateral cochlear implants. Later in the episode, Matt tells us what bilateral cochlear implants do well, and what he's hoping to see from them in the future.

But first, we hear from symposium speaker Brian Monson.

Brian is an associate professor and the director of graduate studies in the Department of Speech and Hearing Science at the University of Illinois, Urbana-Champaign. And he's the lab director and principal investigator of the Auditory Neuro Experience Lab.

As a part of the 2025 ASHA Research Symposium on Hearing, Brian will be speaking on extended high frequencies—those frequencies higher than 8 kilohertz (kHz). He says that range of hearing isn't usually included in an audiogram, even though human hearing ranges from 20 hertz (Hz) to about 20 kilohertz (kHz).

Monson says audiologists test up to 8 kHz in a typical clinical audiogram, and it's an exception for an audiologist to test a higher frequency.

MONSON:

The reason we are studying that higher frequency range is because we would like to know: Why do humans hear in this frequency range?

The implicit assumption of a clinician not testing this frequency range of hearing is that it's not crucial for day-to-day living. Otherwise, they would want to test [it], and we'd want to know how well someone can hear above 8 kHz.

But that assumption seems counter to biology, where if it weren't useful to hear in this frequency range, then humans wouldn't have this frequency range of hearing.

If hearing above 8 kHz weren't relevant for the acoustic signals that humans hear, then there would be no reason to dedicate biological resources to being able to hear above 8 kHz. And that clinical assumption, again—its implicit assumption—runs counter to what biology might dictate. That's why we're studying extended high frequencies, to really ask why is it that we can hear these higher frequencies? What benefit do they provide to someone to navigate their environment or understand speech? That sort of thing.

GRAY:

I know you've written about the relationship between extended high frequencies and speech-in-noise recognition. What are you learning about the relationship between speech-in-noise recognition and extended high frequencies?

MONSON:

What we've discovered is that if someone is able to hear these extended high frequencies that are available in the speech signal, they can provide a benefit for someone who is trying to comprehend speech-in-noise.

We run experiments where we will artificially reduce or remove the extended high frequency range from a sound—a speech sound, a sentence, or recording that we present to someone.

When we remove those extended high frequencies artificially, what we find is that our listeners perform more poorly on speech comprehension in noise. There are other aspects of speech perception that are also affected by loss of audibility of these extended high frequencies.

For example, let's say you're at a party—the quintessential cocktail party, like we like to study in auditory neuroscience. There's someone looking at you and talking to you. And [also] someone who's next to you, but they're facing away from you and talking to someone else.

Then, suddenly, that person turns their head and looks at you and starts talking to you.

That ability to detect the person's head orientation—whether they're facing you or facing away—is an ability that is facilitated by these extended high frequencies.

If someone is able to hear those extended high frequencies, they can more readily determine or perceive if someone has turned their head, to look toward that individual.

GRAY:

What I'm hearing you say is that there's information in these extended high frequencies that people are perceiving, both related to speech perception, but also related to a spatial component about where sound is coming from.

MONSON:

Yeah, that's right.

Now, I like to think of spatial awareness as having multiple facets.

Just one facet of spatial awareness is detecting where a sound is coming from—the location of the sound.

But a different facet, or aspect, of spatial awareness would be to determine the physical orientation of a sound source. That's separate from where the sound is located, but rather, "Is that sound source facing toward me, or facing a different direction?"

GRAY:

Yes. And that's valuable information as a part of communication.

MONSON:

Right. And there's even a third aspect that I might highlight, which would be distance.

Now—this is related, of course, to the sound source location—but a lot of times when we think about sound source location, we might think, "Is it to the left of me? Is it to the right of me?"

Another aspect of that is: "Is the sound source near to me, or is it far away from me?"

And we're digging into that piece right now, to see if these extended high frequencies are valuable in determining whether a sound source is near or far.

So far, our data are indicating that yes, the extended high frequencies can facilitate that ability, as well—to detect whether the sound source is close to you or far away from you.

GRAY (Narration):

Brian says that for decades, it was presumed that extended high frequencies were not valuable for the perception of speech. And then he took me on what he called a speech-science history

lesson.

MONSON:

If you dig through the speech-science literature, what you find is that with the advent of telephony in the late 19th century and early 20th century, there's an uptick in speech science research—driven by the need to transmit speech signals over long distances over telephone wires.

The idea was that it's going to be simpler and less expensive if we can shrink down the amount of information we have to send over these long telephone wires.

The objective became: How much information can we reduce? What kind of information reduction can we do to the speech signal and still be able to transmit something that, on the other end, is an intelligible speech signal?

What the scientists did back then was—they started pulling out different frequency ranges.

They got rid of extended high frequencies and tested whether the speech was still intelligible.

Sure enough, it seemed to have negligible effect if you removed everything above 7 kHz, about.

Now, that's not the only piece. They would also remove lower frequencies—energy below, say, 100 to 300 Hz—and found that you could still reproduce a highly intelligible signal without those low frequencies too.

This led to the telephone bandwidth, which for a long time hung around about 300 Hz to 3,500 or 4,000 Hz.

[If] you remove everything above 4,000 and everything below about 300 Hz, you can still have an intelligible signal. That's good from an engineering perspective, and [for] being able to transmit these signals—because it's cheaper to reduce the information that you have to transmit.

But that, sort of, had a byproduct which became, "Oh, that's the frequency range that's important for speech."

And then it, sort of, became an assumption that these high frequencies just weren't relevant for speech at all.

As late as the early 2000s, you would see studies that were examining the acoustics of the speech signal, and they would present a graph of how much energy at each frequency you see. But the line would always stop at 5 kHz or 6 kHz, as though there was nothing in the signal beyond that—because they just didn't measure it.

There was no interest in that. The idea sort of became: Whatever the relevance of extended high frequency hearing is for humans, we know it must not be speech, right? Because we can low pass filter speech—this artificial reduction or removal of these higher frequencies. We can get rid of these higher frequencies, and it does not affect someone's ability to comprehend the speech signal.

However, those tests were often done in quiet, or with an artificial noise setup, or paradigm that didn't reflect the ecological settings where someone would find themselves in their daily lives.

My observation of how the research had gone is what led me to say, "What if we actually set up more ecologically relevant environments—speech perception environments? Would these extended high frequencies then become useful for speech intelligibility, for speech perception, [and], as you've mentioned, spatial awareness?"

That's really what got me started on this line of research.

GRAY:

And I would wonder about how much the technology was influencing the ability to differentiate sound at that time, as well.

MONSON:

Yeah, great point.

There certainly were technological limitations, which is related to why they wanted to try to shrink down the signal in the first place, perhaps.

But yes, it is technologically more challenging to try to produce a transducer, a microphone, or a loudspeaker that will be able to record or reproduce these frequencies with high fidelity.

There certainly were technological limitations, at the time, that we need to be able to appreciate. They were limited somewhat in their ability to even measure these frequencies in the speech signal, because it is the case that the frequencies above 8 kHz—extended high frequencies—in the speech signal are relatively lower in level. They're not as loud, compared to the lower frequencies that people had discovered were important for speech intelligibility.

There were multiple factors that likely contributed to extended high frequencies for speech sort of being put on the back shelf.

GRAY:

I'm going to ask, in just a second, to connect it back to audiologists.

But before I do, is there anything else that you're learning about extended high frequencies and how people perceive them, and what we do with information we receive from those

signals?

MONSON:

One of the areas that we have explored, and we're starting up a project now to continue to explore how extended high frequencies are beneficial for children.

Most of the work that we have done, and most of the work that's been done in extended high frequency hearing for speech—I would say in general—has been studies conducted with adults.

We have a couple of studies where we have looked at whether children can use these extended high frequencies. And the hypothesis there is that—you recall that I said that you start to lose these extended high frequencies as you age, and that loss at the highest frequencies begins as early as adolescence? That means that children have superior sensitivity at these extended high frequencies.

One hypothesis that you might derive from that observation is that these extended high frequencies are more useful or more important for children—say, to learn speech, or to learn about their environment—than for adults.

Maybe as a child is growing up, they really need these extended high frequencies to help them learn to navigate their environment, to learn about the acoustics of their environment, to learn speech, [and] learn language, perhaps.

And then once they've gotten to a point, say, in adolescence or a little bit older, where they've navigated that, they've learned a lot, and now, maybe the extended of high frequencies become less valuable.

And so, what we call age-related hearing loss, maybe that's a design of the system even to say—we no longer need to dedicate resources to these highest frequencies. We can let them go, if you will.

That's sort of a crude portrayal, perhaps, of what is actually happening, but it's an interesting thought.

In any case, I'm excited and interested to see how the ability to use these extended high frequency cues and their relation to spatial awareness, head orientation of a talker, or distance—how that develops over the child's lifetime.

GRAY:

Early in this conversation, I asked you about audiologists' relationship to these extended high frequencies. What do you think what you've learned means for audiologists? Is there anything that you think could change in the future?

MONSON:

One of the challenges of what we call “research-to-practice,” translational research, and trying to translate the findings that we have in the laboratory to an adjustment in clinical practice is that audiologists have so much that they need to test already.

When someone comes into their clinic, they have a standard battery that takes a reasonable amount of time to get through. To give them information they need to assess the hearing of the patient who has walked in.

As auditory science researchers, many of us—perhaps most of us—are assessing tools that we think could be useful for clinical assessment. We’re setting up studies and testing a paradigm that we think, “Well, maybe this will be useful for implementation in the clinic to identify individuals who have hearing loss, or [are] having hearing difficulties of some sort.”

I’m in that boat. I’m hoping that the extended high frequency work that we’re doing might be able to translate to clinical practice.

One of the findings that we have seen—I would say relatively consistently—is that when you go to the audiology clinic, you get the standard audiogram. What we’ve found is that with our listeners, we test their hearing at 16 kHz. This is higher than the highest frequency that’s typically tested in the audiology clinic.

And that measure, the 16 kHz sensitivity measure, is predictive of listeners’ ability to perform these speech perception tasks that we’re testing them on. If that continues to hold true, that has what we hope to be a relatively simple and potentially pretty immediate clinical implementation that an audiologist can employ.

The audiologist is already doing this tone testing up to 8 kHz, and they have a set of frequencies they do.

If all they have to do is add one more frequency for each ear—in other words, just tack on the 16 kHz tone to your regular battery that [currently] goes up to 8 kHz, that’s going to be a simple addition that’s going to take relatively little time. The listener is there and ready to be tested at that frequency.

Now, that is an oversimplification because not all audiometers—we’re coming back to the technological issue—not all of the equipment that you find in every audiology clinic is equipped to test frequencies higher than 8 kHz.

There’s certainly an adjustment that would need to be made there. But in terms of the time that it would take, and there’s no additional training needed for the audiologist; it is something that they are trained to do already.

That’s one clinical implementation that we think is feasible, and I would hope that audiologists would find useful in identifying individuals who may be at risk, for example, of listening difficulties, particularly in noise.

GRAY:

Anything else that you wanted to share, or that you thought I would ask about that I didn't?

MONSON:

When you're wanting to conduct these tests of extended high frequencies, you have to make sure that your loudspeakers, your microphones, and any recordings of speech that you're using have the extended high frequencies in them, or are able to reproduce the extended high frequencies with high fidelity.

That's an important thing to think about because, again, over the history of speech research, the thought was that the extended high frequencies aren't very useful. And so many speech recordings made during that time—I'd say most—didn't bother capturing the extended high frequencies in the recordings.

One thing that we are working on, to rectify that, is to create high-fidelity, full-bandwidth speech recordings. We do have a publicly available corpus of these high-fidelity speech signals. And if someone were interested in that, they could just Google my name and could contact me, and I could point them in that direction.

Thank You

GRAY:

Brian, thank you so much for sharing this with me today. I appreciate it.

MONSON:

Absolutely. It's been a pleasure. Thank you, J.D.

GRAY (Narration):

Brian told me he wants to acknowledge the National Institutes of Health and National Institute on Deafness and Other Communication Disorders, who are funding his work.

We continue our conversations on research with Matt Goupell.

Matt is a professor at the University of Maryland, where he is the director of the PhD program and the co-director of undergraduate honors for the Department of Hearing and Speech Sciences.

He's also a co-director of the Center for the Comparative Evolutionary Biology of Hearing.

At the Research Symposium on Hearing, Matt's presentation is titled, "Improving Hearing with Two Ears in Adults with Cochlear Implants."

I started by asking Matt: “What do bilateral cochlear implants do well, and what are you hoping they might improve?”

GOUPELL:

Well, two ears are better than one.

And the reason that two ears are better than one—and that we as human beings and most mammals have two ears—is that it helps us tell where sounds come from. The direction of a sound.

That’s important if you want to know where things are coming from. Maybe, crossing the street, or if you want to run away from some predator.

It’s also helpful for understanding speech in noisy situations.

You might be trying to pay attention to somebody at a loud party or loud restaurant, and there’s sound coming from every direction.

Having two ears allows you to focus better on the person you’re trying to talk to and turn down the sounds from every other direction. You get better at that when you have cochlear implants in both ears.

But they’re not perfect, right?

You’re not completely restoring the typical perception. There are still challenges when you get into these situations.

A lot of my research helps try to improve spatial hearing with cochlear implants.

GRAY:

Tell me a little bit about what you’re learning about spatial hearing for people using bilateral cochlear implants.

GOUPELL:

The way that we process spatial information in the brain—we have our two ears. You can kind of think of them as two microphones, and they’re getting these signals. And then you have to compute the differences between the ears to actually tell where sounds are coming from.

You get timing differences between the ears, and you get level differences, or loudness differences between the ears, depending on where the sound is coming from. And that helps your brain determine the location of the different sounds in space.

Bilateral cochlear implants do a much better job of conveying level-information across the ears than the timing information. The timing information is not well-encoded because the devices aren’t synchronized.

Because of that, we've lost one of these important cues. And, unfortunately for human beings, the timing differences are really the main localization cue.

We as human beings are so good at localizing sounds, left-right, because of timing cues. And by losing those, that's a huge hit to our ability to separate and segregate sounds, and tell where they're coming from.

GRAY:

I heard you say, here's kind of a benefit of bilateral cochlear implants: [they] can help you with spatial awareness, but the timing isn't maybe what it could be.

What are some of the real-world effects of the timing part of it?

GOUPELL:

The real-world effect here is that instead of being able to precisely point in the direction of a sound, you might be able to say, "Well, that sound is to the right of me, or it's to the left of me." But you can't really say, "Oh, well, it's a little bit to the right, a little bit to the left," [or] exactly where it is.

I guess one way to think about it is if you're searching for a sound in space, the nice thing about having two implants is that you don't necessarily have to turn all the way around in a circle to see where the sound is coming from, but you're going to have to turn up to 180 degrees to the left, or up to 180 degrees to the right. And that's just because we've lost one of these important cues—these interaural time differences.

GRAY:

Are we currently seeing an increase in the use of bilateral cochlear implants?

GOUPELL:

That's a good question.

I'm going to take a guess. I actually don't know the numbers off the top of my head, but more and more people are getting cochlear implants.

And, for example, many kids are getting implants at a very early age if it's appropriate. And when they do, it's often that they're getting bilateral cochlear implants.

That's pretty standard-of-care these days. Adults also will get implants, adults who may have had hearing as a child and then lost it for some reason as they get older. And in those situations, they might get bilateral implants. They might get one implant.

But in general, I think that more and more people do have bilateral implants. And the reason is that there's just—in the end, there is an advantage to having two ears, and that's going to help with these two primary spatial hearing tasks.

There are, of course, caveats and exceptions—it's not appropriate for all people—but broadly, the benefits are there. It's just not going to get you to the point where you're doing as well as someone with sort of typical acoustic hearing. And that's the real goal, is how can we get people to do better with the localization, and how do we get people to operate in these noisy situations like restaurants where they're not struggling as much?

GRAY:

And that may be something that an audiologist will be able to help with.

GOUPELL:

Well, that's the hope. That's the hope.

I mean, there are lots of things that we would like to be able to do, and we're looking for the tools to do it.

Some of it's going to come in terms of the companies making changes to the devices, synchronizing the devices at some level, so that they are encoding the time differences better. That would be one manufacturer change.

But all of these interaural differences are based on symmetry, or matching, between the ears.

That's something that an audiologist can have access to when they program an implant. You might be doing some loudness balancing across the ears to make sure they're about the same loudness. You could possibly do that at a single electrode level, because cochlear implants have multiple electrodes that are providing frequency-specific information.

There are now more and more people getting CT scans after cochlear implantation. And when you have a CT scan where the devices are ... For example, if an implant is in one ear deeper than the other ear, you could correct for that and line up the frequency information across the two ears. And that, potentially, could be a benefit too.

We think that matching the two ears—making them as symmetric as possible—should conceptually help you hear better, in terms of spatial hearing.

GRAY:

What else are you learning about bilateral cochlear implants?

GOUPELL:

I mean, golly, I just brought that up. And at the same time, these devices are—they don't have very specific frequency information, and there's all this current spread. Each electrode excites such a wide range of neurons, and such a wide range of frequencies.

You actually need a pretty big difference between the insertion depths between the ears to have a negative consequence that you might need to fix.

And one thing that I learned—actually, from looking at a bunch of CT scans—is that a lot of people are actually more matched across the ears and insertion depth than I realized.

I thought this was a big problem, and maybe it isn't. Given that, I'm turning my attention more to thinking about if I don't want to fix the frequency mismatch problem, then I should be spending more time and effort with loudness balancing across the ears, and fixing the timing issues between the ears.

GRAY:

It's interesting to think about where maybe [cochlear implants] originated—where they were before, and where they're headed next.

ASHA is turning a hundred this year. We've been talking a lot, on the podcast and in other programming, about the future.

I'm wondering—maybe not a hundred years from now, but what do you hope to see in the future for cochlear implants?

GOUPELL:

I do think that there are probably ways that we can convey more of the important spatial cues.

I mean, synchronization of the devices across the ears is going to be a critical step, and trying to get more frequency specific information is also going to be important too—into fixing some of our main problems.

If we could fix those couple things, I think that that would be a big step in helping people with spatial hearing. But it'd probably help in a great number of things too.

Getting better frequency selectivity would help in other domains, like pitch.

And pitch is another good cue to help separate sounds out when there are lots of sounds going on the same time. It would help you appreciate music too.

Cochlear implants just don't convey pitch and music well right now either, and it's a big stumbling block. I'm thinking that if we can get those two things fixed, moving forward, that's going to be the next big leap forward in the technology.

GRAY:

Matt, I've noticed throughout our conversation you referenced the acoustics and the environment. You've referenced electricity and computers, and also individuals' experience of sound.

I know that you have a multidisciplinary approach to this research. I wonder if you could just talk briefly about what all those different elements bring to your work.

GOUPELL:

Yeah. Yeah. What I usually don't lead conversations with—at parties—is that my background is in physics and math. It gets people to turn around right away.

Yeah, that's where I come from—the physics and the math.

The physics is important, because when you think about spatial hearing, you think about how sounds are being generated by a source, how they're traveling from that source to someone's ears, and then the whole transformation of that sound, and all of the processes that end up getting encoded into little electrical pulses in the brain.

I think a lot about the physics of these situations, and the physics are messy. The physics are complicated.

And the brain is messy. The brain is complicated.

And so are implants.

Maybe that's a good thing, but it's good to have these different perspectives.

I came to this field studying hearing and auditory neuroscience, just because it was such a great interdisciplinary marriage.

You have all of these different people with all these different backgrounds, and it was one of the few places where you saw people with physics and engineering backgrounds interacting with audiologists and clinicians and surgeons. But also you had people from psychology and behavior, and they're all kind of mixing together and bringing their own expertise. And it ends up making a fun environment to think about how to help people here, and the different dimensions of what's really going on when we're asking people to do our studies and tests.

GRAY:

Thank you so much for your time today.

GOUPELL:

Thank you.

GRAY (Narration):

You can hear more from Matt Goupell and Brian Monson at the Research Symposium on Hearing at the 2025 ASHA Convention in Washington, D.C. And for those who won't be making it to Convention, you'll be able to find livestreams of sessions from both speakers. You can find details in the show notes for this episode, or on our website.

And if you look to the podcast archive, you can hear episodes featuring past research symposium speakers discussing topics like public health, artificial intelligence, and gene

therapy.

ASHA Voices is produced by the American Speech-Language-Hearing Association and comes from the team behind the ASHA Leader magazine.

I'm J.D. Gray. This is ASHA Voices.



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